Experiment Name:

HTTP Web and FTP Server Access with DNS using Cisco Packet Tracer (Simulation)

Introduction:

Computer networks use services such as HTTP, FTP, and DNS to enable communication and resource sharing. HTTP is used to access web pages, FTP is used for file transfer, and DNS resolves domain names into IP addresses. In this experiment, all network components and services were simulated using Cisco Packet Tracer. No physical hardware was used. The goal was to configure and test HTTP and FTP server access using DNS in a virtual network environment.

Software and Tools Used:

1. Cisco Packet Tracer (Network Simulation Software)
2. Simulated PC (Client)
3. Simulated Router
4. Simulated Switch
5. Simulated HTTP Server
6. Simulated FTP Server
7. Simulated DNS Server

Procedure:

1.Open Cisco Packet Tracer.
2.Add simulated PCs, router, switch, and servers to the workspace.
3.Connect all devices using virtual Ethernet connections.
4.Assign IP addresses, subnet masks, and default gateways to all simulated devices.
5.Open the DNS server in Packet Tracer.
6.Turn the DNS service ON.
7.Create a DNS record for the HTTP server domain name.
8.Create a DNS record for the FTP server domain name.
9.Save the DNS server configuration.
10.Open the HTTP server device.
11.Enable the HTTP service.
12.Verify the default web page.
13.Open the FTP server device.
14.Enable the FTP service.
15.Create a user account for FTP access.
16.Configure the DNS server IP address on the client PC.
17.Use the web browser to access the HTTP server using the domain name.

18.Use the command prompt to access the FTP server using its domain name and verify successful communication.

Conclusion:

This experiment was successfully completed using software simulation only in Cisco Packet Tracer. It helped in understanding how HTTP, FTP, and DNS services work together in a network without using real hardware. The experiment provided clear insight into server configuration and domain-based service access in a simulated networking environment.